

Ch1

Ch2

Ch3

Ch4

Ch5

Ch6

Welcome

Lit. Review

Methodology

Design

Results

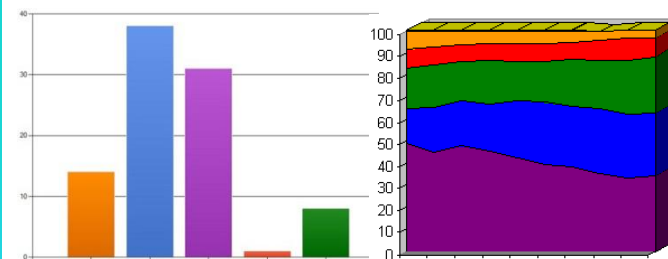
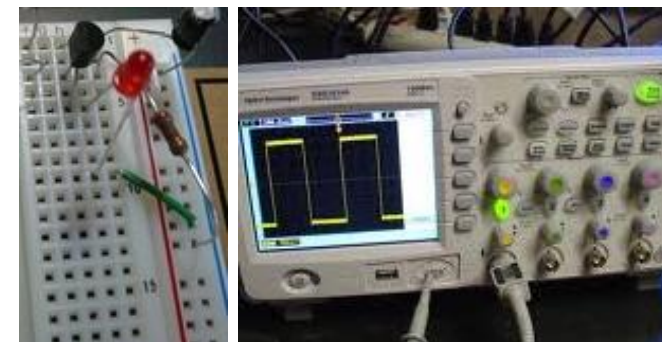
Introduction &
Conclusions

Experiment Refinement & Documenting Results

5th part of the guided research track (GRT)

RRSG Seminar

Prepared by Dr. Simon Winberg



Today's Meeting plan

Make sure to
look at this!

- The marking schema – planning the write-up
- Short presentation (around 20mins):
 - Refining your experiment plans
 - Considerations for writing the Results Chapter(s).
- Snippets of results imagery from past years (mostly copied from past orals presentations)
- View of results chapters from past years
- Discussion of design issues
- Q&A

[See last slide for references.](#) These may also be useful to help you prepare and write your results chapter.

Where you should be in your project

- You should already be **past stage of system level / high level design**
- You should **be well into the implementation** and ideally already **generating some results**

You should be...



well into the implementation phase
and hopefully have done some initial testing at this stage.

GRT

THOUGHTS ABOUT GETTING HIGH MARKS

DEPARTMENT OF ELECTRICAL ENGINEERING
FINAL YEAR PROJECT REPORT MARKING SCHEDULE - August 16

Student name:	Student No:
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Report Title:

Mark allocation should be moderated by the difficulty of the project set.

	Weight (%)	Comment Examiner writes brief comments here; adequate to justify mark. Use separate page if needed.	Assign a % mark out of 100% in each of the five boxes
Problem statement & literature review:	20		
Understanding the nature of the investigation			
Acquaintance with relevant literature			
Design and Theory	20		
Practical work, simulation and software	30		
Results: Interpretation and conclusions	20		
Presentation, layout, referencing	10		
TOTAL	100	Report Mark (%)	

CH1

CH2

CH2 & refs elsewhere

CH3&4

Methodology & Design

CH5&6

Results & Conclusion

All

Consistency in quality

playing the rubric...



Or more accurately, getting your report to align perfectly with the marking rubric

The marking rubric

EEE4022F/S Final Year Project – Rubric							
Criteria	0-39 %	40 to 49% Unacceptable	50% to 59% Acceptable	60% to 69% Good	70% to 74% Very Good	75% to 85% Excellent	86% to 100% Outstanding
a) Problem statement & literature review: Was the research question / engineering problem clearly formulated and achievable? Did the student understand the nature of the investigation? Is the literature covered, relevant, thorough, pertinent, of good quality and suitably extensive?	M i s s i n g o r t o t a l l y u n a c c e p t a b l e	Poorly formulated research question/ engineering problem. Fragmented, incomplete and unclear research question/engineering problem. Superficial discussion and/or no academic sources.	Not completely clear, but reasonable attempt at formulating achievable question/ problem. Relevance and engagement in part but remains unfocused, unclear link to question/problem, literature missing and/or unreliable.	Good question/problem formulation but verbose and beyond scope of report. Reasonably relevant, linked to question/problem, some important literature covered.	Clear question/problem formulation but perhaps not achievable within the report scope. Relevant, linked to question/problem, majority of important literature covered. Clear and achievable question/problem concisely stated.	Mastery in question/problem formulation - strongly crafted within achievable scope of report. Strong relevance linking literature to question/problem and comprehensive coverage of the important literature.	Original, engaging, and thought provoking research question/engineering problem. Critical engagement with literature. Relevant, supports question/problem, covers all aspects, credible sources used.
b) Design and Theory Is there clear development of theory? Is the design well described? Is the description clear enough to allow another student to duplicate it? Does the understanding of the theory inform the design of the project.		Very little understanding, poor application of theory.	Some discussion of method and limitations. Incomplete theory development and design description.	Partial discussion of method and limitations, some theory development. Weak design description.	Satisfactory discussion of method, some limitations, and justification for using method in terms of question/problem. Solid design description	Thorough discussion of method and limitations. Good theory development and complete design description for replication.	Creative development of methodology/theory which progresses method in the field. Meticulous care in design description.
c) Practical work, simulation and software How much practical work has been done? How relevant was it to the topic? Were the procedures verified, either by experiments or clear reasoning?		Inadequate practical work done i.t.o. quantity and quality.	Barely adequate in terms of either quantity or quality, some useful content.	Adequate in terms of quantity and/or quality but fails to address questions/problem fully.	Suitable practical work done which addresses research questions convincingly.	Thorough practical work done which fully covers question /problem.	Exceptional rigour demonstrated in practical work.
d) Results: Interpretation and conclusions Are the results interpreted clearly? Are the conclusions reached clearly supported in the results and linked to the research question/engineering problem? Are suggestions made for further research?	a c a d e m i c l e v e	Superficial and/or irrelevant analysis. Little or no synthesis of information, no link to research questions/problem, no suggestions for future research.	Little synthesis of information, some interpretation, but poor link to research questions.	Some interpretation which partially answers research question/engineering problem. Some weakness in summary of results and suggestions for future research.	Good attempt, but some conclusions absent or not fully supported. Adequate analysis which covers basics of research questions/engineering problem. Some suggestions for further research/work.	Accurate analysis, drawing out features of the data and speaking to research questions/engineering problem. Good suggestions for further research.	Creative analysis which draws out pertinent solutions to research questions /problem & highlights additional features. Succinct and precise conclusions. Useful and creative suggestions for future research offered.
f) Presentation, layout, referencing		Unprofessional work. Poor	Problems with layout, careless grammar	Layout and presentation could be	Layout and presentation acceptable. A few minor	Very few, if any, errors in citations. Care	No errors in referencing and grammar.

Let's zoom in on the top categories

What will boost your EEE4022
into the stratosphere & beyond?



Excellent / Outstanding Problem Statement

- Mastery in question/problem formulation - strongly crafted within achievable scope of report.
- Original, engaging, and thought provoking research question/engineering problem.
 - *Main thing is to express clearly the problem statement and research questions, a table of requirements and tests helps with this*

Excellent / Outstanding Literature Review

- Critical engagement with literature.
 - Means the objective *analysis* and *evaluation* of an issue in order to form a judgement.
- Relevant, supports question/problem
 - Clearly connects to objective/questions/problem posed in the introduction
- Covers all aspects
 - Provides a comprehensive response
- Credible sources used
 - Good textbooks, peer reviewed articles from reputable journal/conference

Excellent / Outstanding Design and Theory

- Thorough discussion of method & limitations.
 - *But not a laborious discussion!*
- Good theory development and complete design description for replication
- *Use some diagrams too in the lit review!!*
- Meticulous care in design description.
- For truly exceptional:
 - Creative development of methodology/theory which progresses method in the field

Excellent / Outstanding Practical Work

- Fully covers question / problem.
- For max marks:
 - Exceptional rigour
 - But managing a balance also, in that your 'thorough but concise', i.e. not just dumping all the possible design data and decisions, but rather suitably selecting relevant aspects and sufficiently important (e.g. for another student to repeat the project) details to document
- *Excellent figures, photos, screenshots goes without saying – these are very important*

Excellent / Outstanding

Results, Interpretation, Conclusions

- Accurate analysis drawing out features of the data and speaking to research questions/problem.
- *Good and effective use of graphs/illustrations*
- Highlights additional features observed in experiments.
- Clear and thorough explanations.
- *Refer to your figures, don't just assume the reader will interpret your visuals*
- Good suggestions for further research.

Excellent / Outstanding Presentation, Layout, Referencing

- Care taken in the writing
 - *Well constructed sentences!!*
 - *Leave no figure unmentioned!!*
- No spelling and grammar mistakes
- Consistent referencing style
 - *Use a reference manager like Mendeley!*



Get all that sorted & you're beyond the stratosphere in terms of EEE4022 marks

Let's think about the
RESULTS
chapter

Results Chapter

- Your results should be clearly aligned to the methodology (CH3) where you describe how you are going to test the system and obtain results from the testing.
- It should be logically structured, following an effective sequence, e.g.
 - Initial tests (possibly even phase 0, pre-study results)
 - Progressively more comprehensive testing

Results Chapter

- Unsure of what to show?...
 - *Err on the side of more rather than less results*
- How so?
 - Don't just give the final testing results, unless it is indeed quite a complex system you are building.
 - Can provide evidence of having follow a research-based or experimental research design in which you test and validate decisions before moving on.
- Be generous with the visuals
- Remember to describe the figures...
do not just throw in lot of figures and not explain and discuss them.

Strive towards a Suitably 'balanced selection of figures and text' ...

Too many figures/data

But too many figures may lead to =>

Results camouflage = the important results are lost in the crowd

Confustication = confound + confuse

Marks:

100%

0%

Confustication level

Lead to poor marks



Principle: Consider each figure worth +/- 1000 words: too much data = information overload.

Important

Don't leave documenting
(or at least recording) &
describing results to the
end!!

Why shouldn't you leave documenting results to the last few days?

Here are some reasons...

The descriptions might end up being of an inadequate quality, i.e. vague / messy or look too much like being done in a rush.

Potential misalignment between objectives -> method -> result (i.e. saying one thing and doing another)

Might not have time to show sufficient results, making the examiner think you didn't do enough practical work.

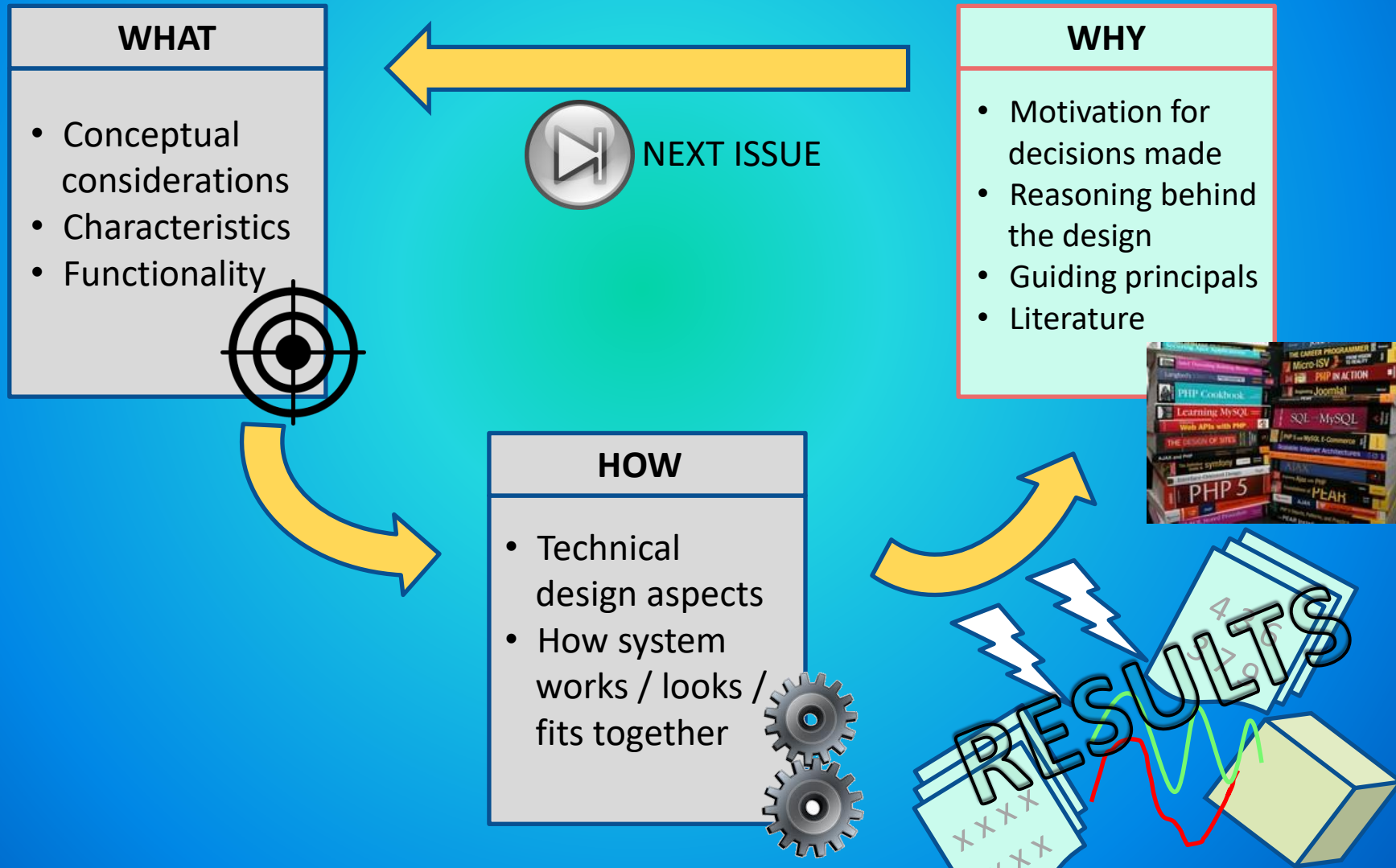
(Highly problematic common problem):

Forgetting what you did, or losing results, and having to redo the experiments (likely done less carefully in the last minute)



Recall the project is likely follows the usual spiral model...

Results within the interactive and iterative Design Process

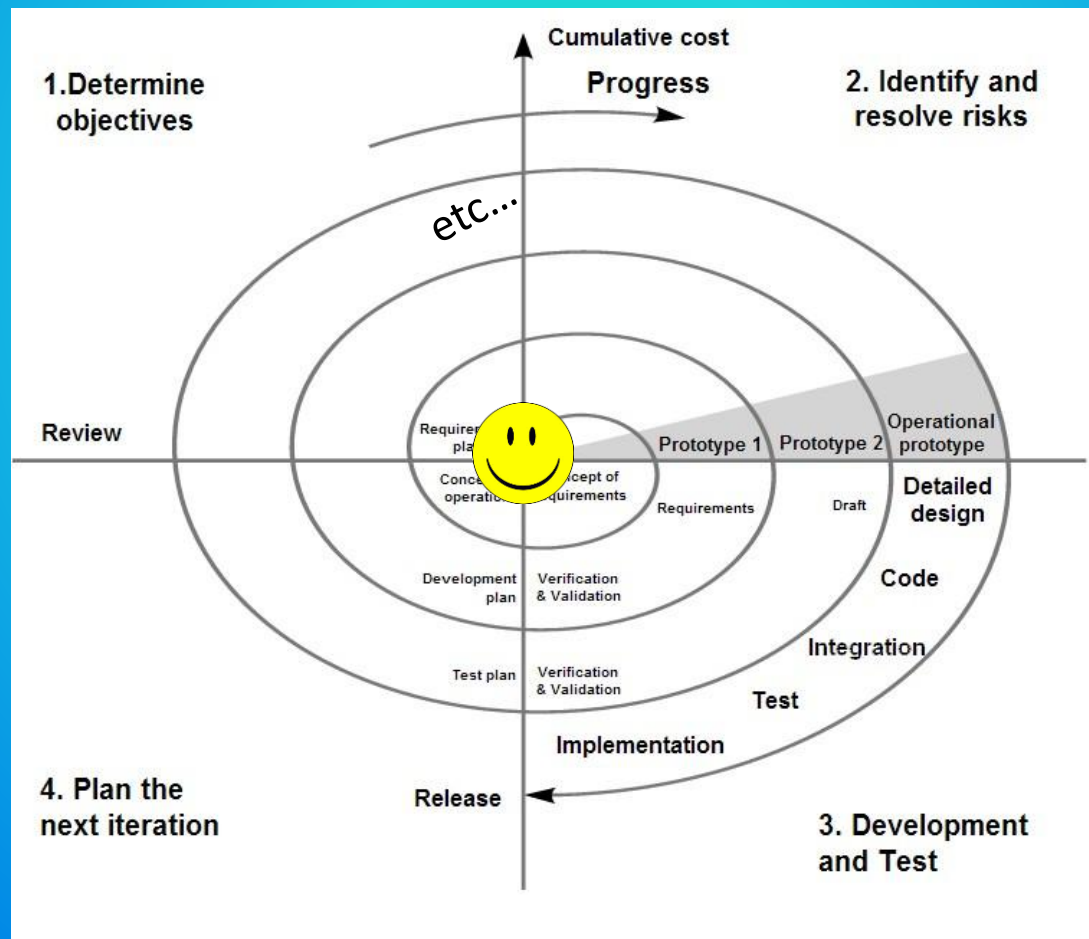


The spiral model of project development

- Remember: Spiral model



Developer



Your Results Chapter

- Overview

Structure of the results chapter is elaborated in the report template

- How the results are structured, brief summary of what was done, refer back to the methodology.
- Subsystem or module tests (possibly prior version tests if it can only be tested as a full system)
- Functional testing; Module testing; Regression testing
- Integration testing
- System testing
- Acceptance testing
- User testing (+ e.g. Surveys)
- [Maintenance testing]* advanced (shows how easy / maintainable the system is)

Alternate Result Chapter(s) Structure

- To a certain extent, what your project involves influences how your results are shown. e.g...

In some cases it may be better to document 'subsystem design – experiment - test' chunks, followed by an integration / full system test i.e.

Chapter 1: Intro
Chapter 2: Literature review
Chapter 3: Methodology
Chapter 4: Subsystem 1
 4.1 Design of subsystem 1
 4.2 Testing / Results of subsystem 1
Chapter 5: Subsystem 2
...
Chapter N: Integration / System Testing
Chapter N+1: Conclusion

Recall: types of testing

- Two broad types of testing:
 - **Functional testing** = testing just the functionality of the application
 - **System testing** = end to end / full product testing, i.e. testing of the whole system ideally in the context that it is meant to be used in.



Functional testing

Typically quicker to do, lab-based



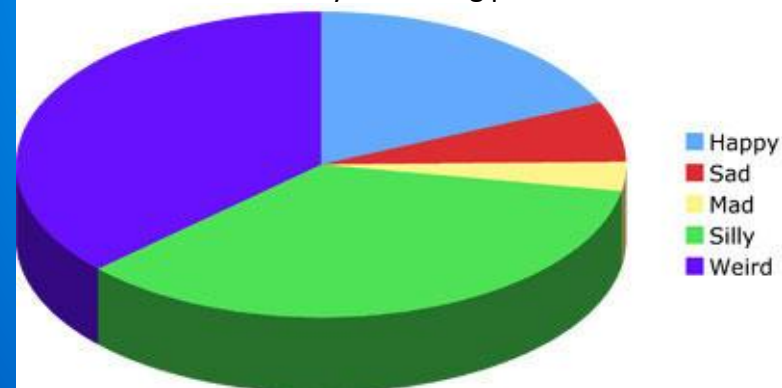
System testing

Typically more involved, onsite

Visuals

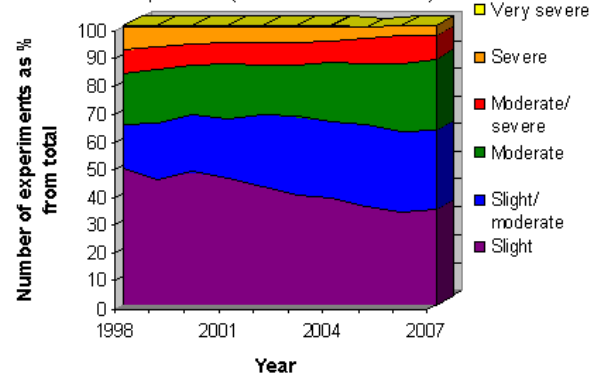
- It's nice to have good use of visuals and tables in the results.
- You don't have to limit yourself to basic line or scatter graphs! Sometimes other graphs or visuals can show the results more effectively.

Fictional study of reading preferences

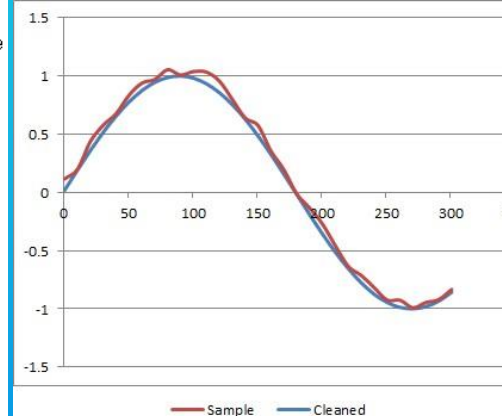


Pie chart

Level of distress in experiments (Netherlands 1998 - 2007)



Stacked area graph



Line graph

Quantitative & Qualitative Results

- Engineers generally expect mostly quantitative result reporting
- But there should be some of both
- But again, it depends on the nature of the project concerned.
- Usually:

More People involved -> more qualitative results expected

Less people involved -> more quantitative results expected

Remembering that:

Qualitative = measurement of quality rather than its quantity.

Quantitative = measurement of quantity rather than its quality.

NB: Your Results should correspond To Your Objectives!

- Try to show good consistency and integration between the chapters of your write-up
 - Remind the user or emphasize how particular results link back to the requirements & main objectives
 - Make use of cross-references to help show the connectedness:

*connect: objectives →
requirements → specifications →
design → testing & results →
conclusions → future work*

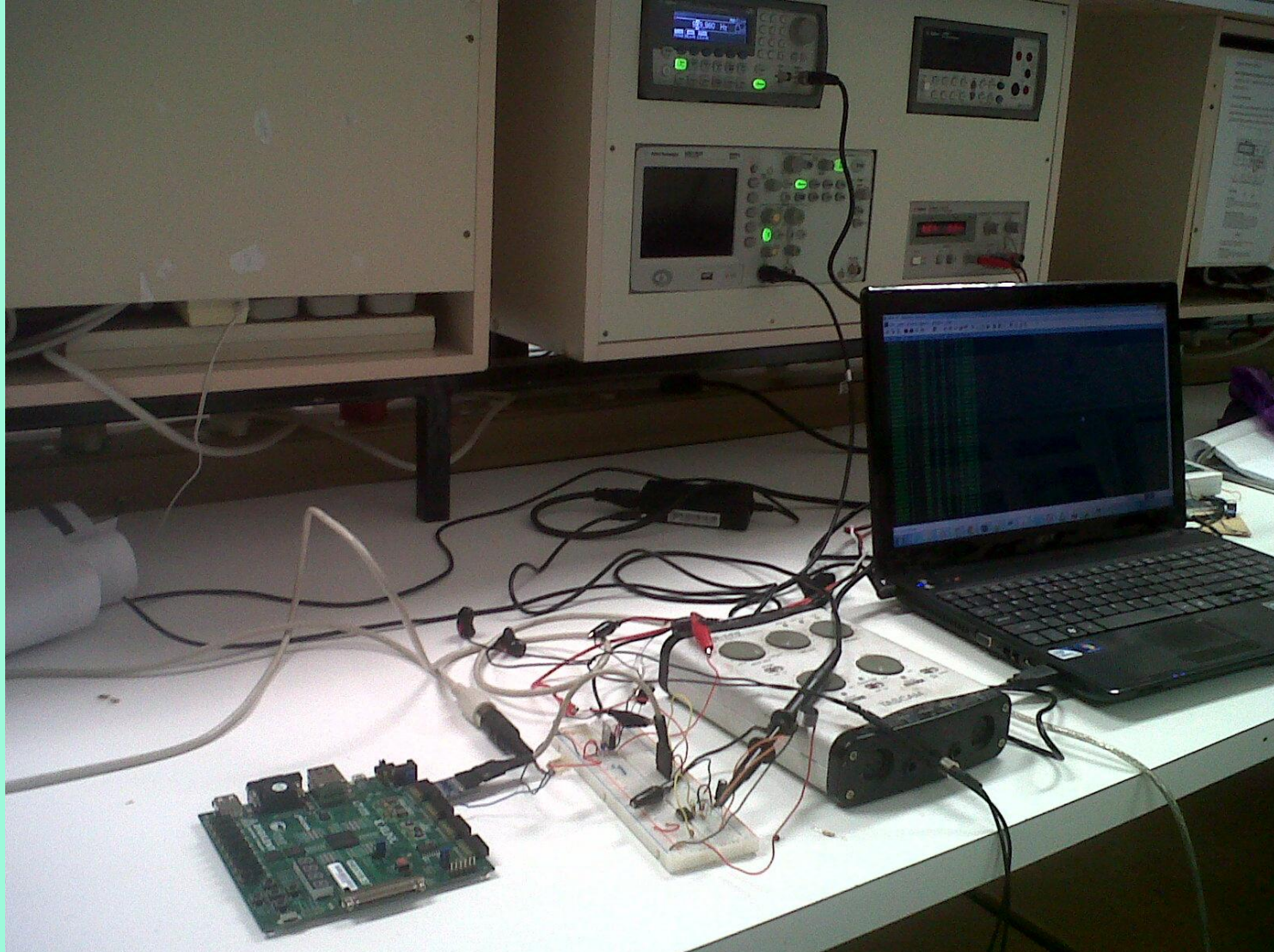
Keep your eye on
the ~~ball~~ objectives



Demonstration & discussion: Results shown in past BSc Orals



A setup snapshot



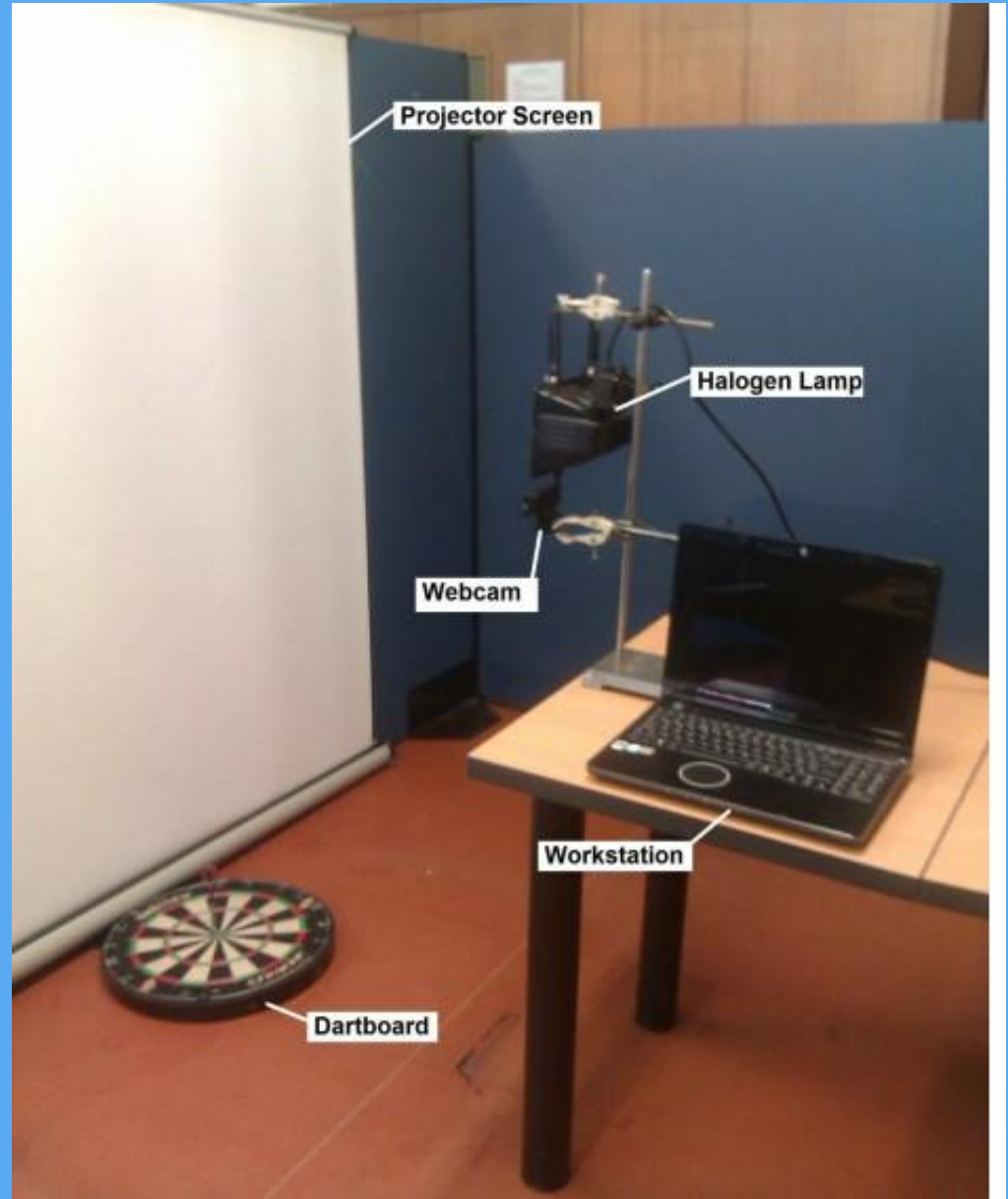
Thabiso Nthathakane -- whistle to Midi convertor system

Experimental setup

Falling Dart Scenario

Components:

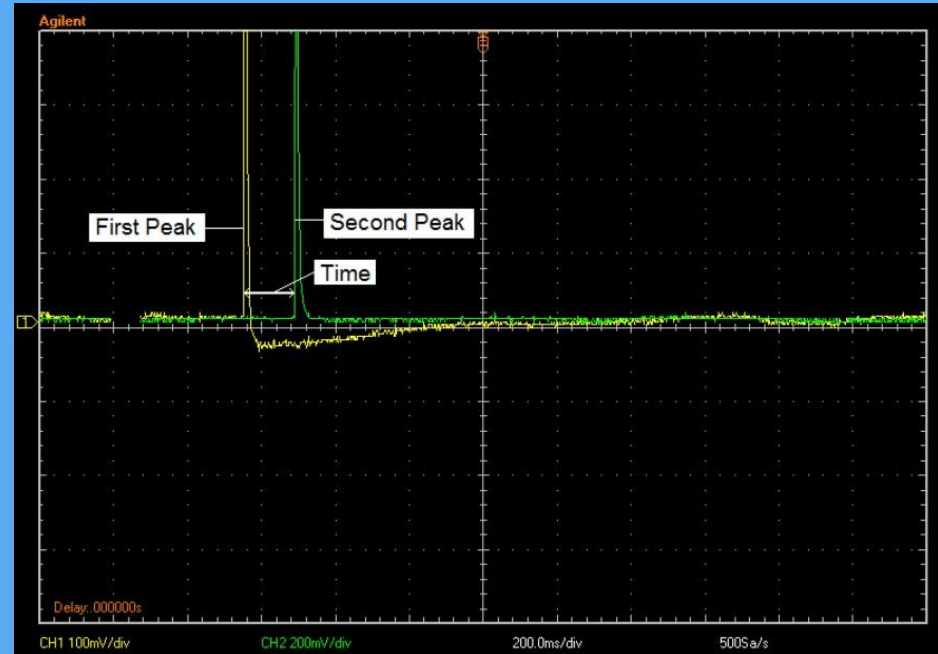
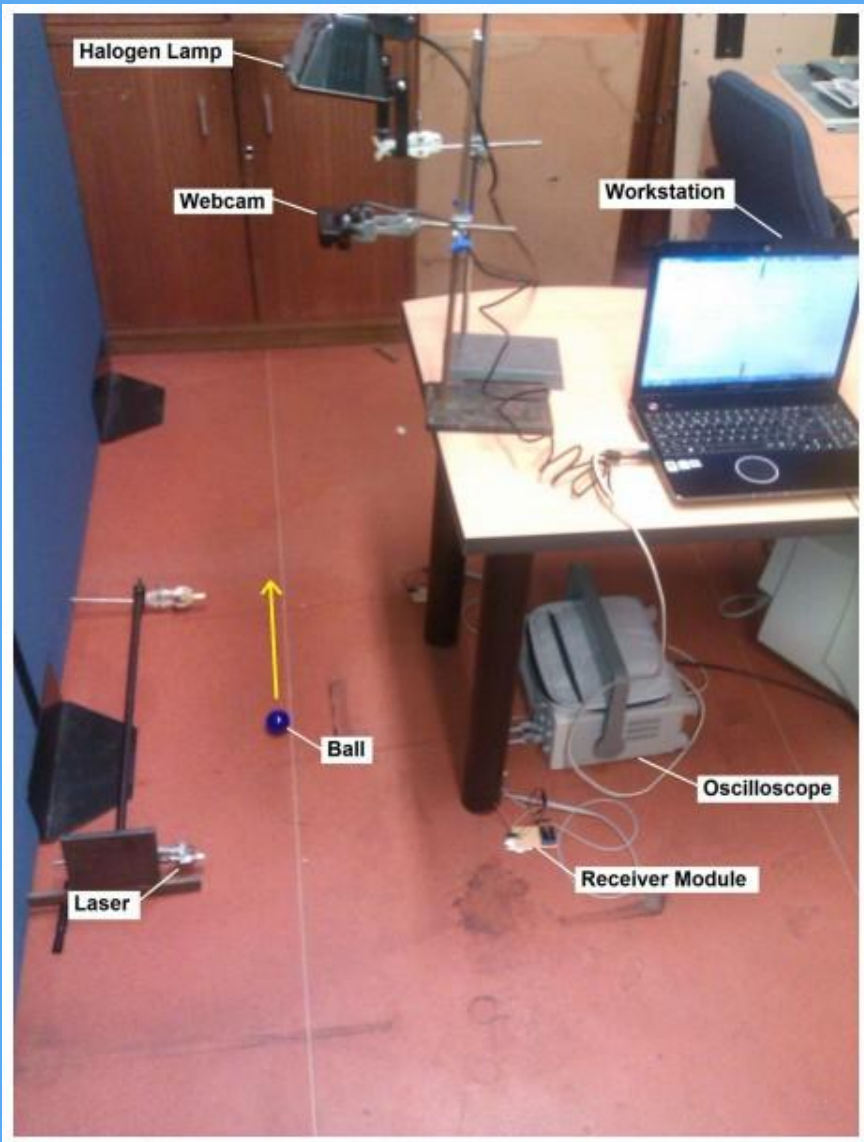
- Workstation
- Webcam
- Halogen lamp
- Projector screen
- Dartboard



“Vision Based Dart Speed Meter” - Matthew Koen

Experimental results

Rolling Ball Scenario



Snapshot of oscilloscope showing time values

The data capturing process was as follows:

1. Firstly, the camera was calibrated
2. User clicked a button to initiate acquisition of images
3. Ball rolled at moderately high speed along the white line
4. The ball crossed the first and then second laser, creating two spikes in the waveform
5. After 3 seconds, image acquisition was terminated
6. Finally, a snapshot of the oscilloscope's display was uploaded to the workstation.

"Vision Based Dart Speed Meter" - Matthew Koen

Review of existing products

“What vendors made”

Functionality/Feature	P1	P2	P3
MASTER STATION			
• Billing System	☺ Yes	☺ Yes	☺ Yes
• Vending System	☹ No	☺ Yes	☺ Yes
• Data Warehouse	☺ Yes	☺ Yes	☺ Yes
• Load Management	☹ No	☺ Yes	☺ Yes
• Tamper Detection	☹ No	☺ Yes	☺ Yes
• Fault Investigation	☹ No	☺ Yes	☺ Yes
• Quality of Supply	☹ No	☹ No	☺ Yes
AMI METER			
• CIU	☹ No	☺ Yes	☺ Yes
• ACD	☹ No	☺ Yes	☺ Yes
• Load Switch	☹ No	☺ Yes	☺ Yes
COMMUNICATION			
GSM/GPRS	☺ Yes	☺ Yes	☺ Yes
PLC	☹ No	☹ No	☹ No
ZigBee	☹ No	☺ Yes	☺ Yes
TCP/IP	☺ Yes	☺ Yes	☺ Yes
PSTN/IDSN	☺ Yes	☹ No	☹ No

Validating Simulation

Aim: To determine utilization of the communication network.

Methods and Materials: OPNET 14.0 used to model the AMI network for The City of Cape Town.

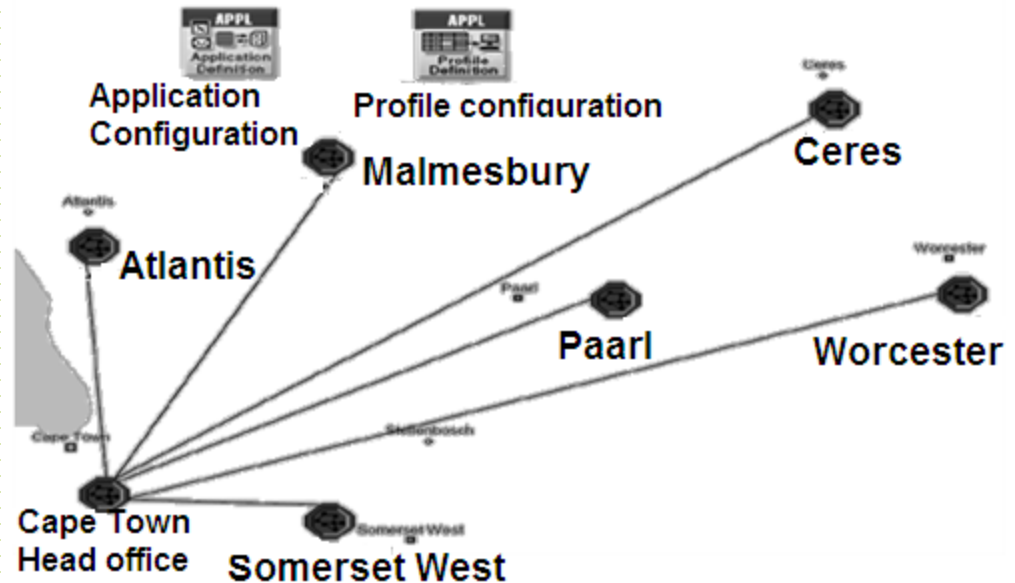
OPNET snapshot of Modelled network:

Legend:



Subnet

The subnet represents a detailed network of AMI meters for a city.



Validating Simulation Results

Graphs 1 and 2 show maximum utilization of network capacity between Cape Town control centre and Atlantis and Ceres respectively

Graph 1:

Link Type: Optic Fibre

Link bandwidth: 100Mbps

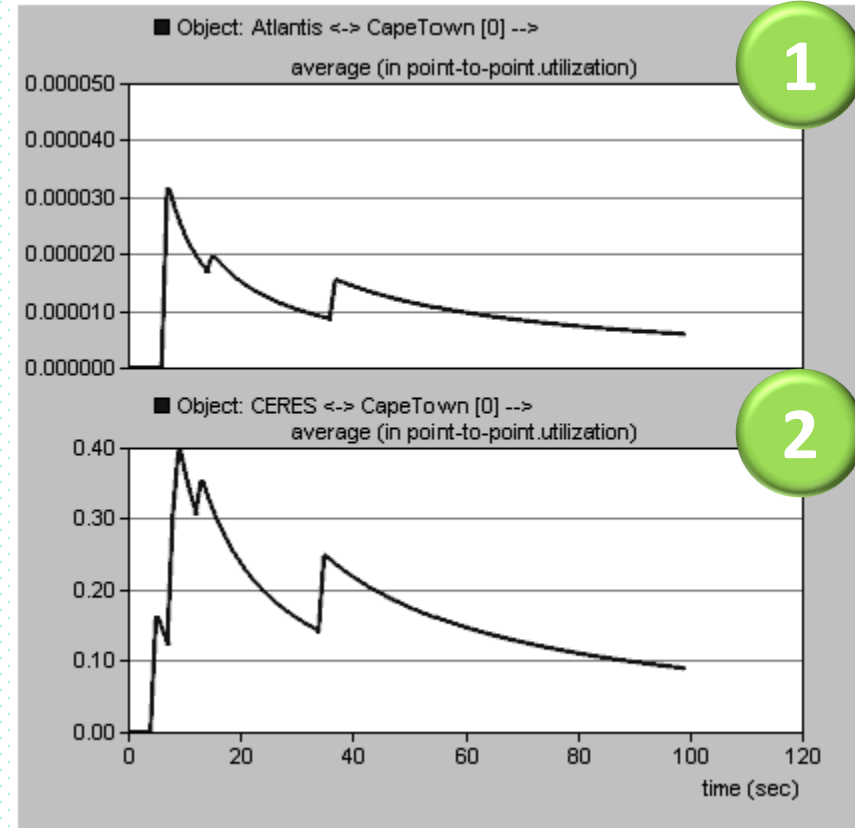
Utilization: 0.032%

Graph 2:

Link Type: DS0

Link bandwidth: 64kbps

Utilization: 4%



Demonstration & discussion: Results Chapters from past BSc reports





Now over to you

Discussion & Questions

Ch1

Ch2

Ch3

Ch4

Ch5

Ch6

Welcome

Lit. Review

Methodology

Design

Results

Introduction &
Conclusions**Planning ahead:**

- No meeting planned for next week
- No meetings for a few weeks until...
- (meeting to be announced): last meeting 1 week before due date;
GRT Presentation on: Conclusions and getting the Introduction to link nicely to the conclusions and rest of the report

Refs

- Some recommended reading...
- **How to write a dissertation:** a free online short course (useful content you can view without doing the course)
 - Course start page: <http://www.mondofacto.com/study-skills/research/how-to-write-your-dissertation/>
 - Writing up Results: <http://www.mondofacto.com/study-skills/research/how-to-write-your-dissertation/04.html>
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